Summer Math Booklet – Complete Answer Key

Grade 4→5 Transition · Peoria, AZ · iReady Aligned · Beast Academy Style

FOR PARENT/GUARDIAN USE: This answer key includes step-by-step solutions. The goal is for your child to learn FROM his mistakes, not just see the answers. After each day's work, have him check his own work using this key, explain any step he doesn't understand, and try a similar problem on his own.

How to Use This Key

1. Child completes the day's page independently.
2. Child checks answers using the key.
3. For wrong answers: child reads the step-by-step solution slowly, then covers it and re-solves the problem.
4. If still confused: parent walks through the example together.
5. Quiz answers are given at the end — do not share these until the quiz is done.

Topics by Week:

- **W1** Number Sense — Place Value, Rounding, Comparing, Expanded Form
- **W2** Fractions — Equivalence, Comparing, Adding/Subtracting, Mixed Numbers
- **W3** Multiplication & Division — Multi-digit, Long Division, Factors, Word Problems
- **W4** Decimals — Place Value, Reading/Writing, Comparing, All Operations
- **W5** Geometry — Angles, Triangles, Quadrilaterals, Area/Perimeter, Coordinates
- **W6** Exponents — Base/Exponent, Powers of 10, PEMDAS, Perfect Squares & √
- **W7** Expressions & Equations — Variables, Like Terms, 1-Step & 2-Step, Inequalities
- **W8** Probability — P Formula, Scale, Experimental/Theoretical, Tree Diagrams, Predictions

Week 6–8 Answer Key Location:

Answers for Weeks 6, 7, and 8 are printed at the end of **summer_booklet_weeks6_8.html**.
Open that file and scroll to the last section titled "Answer Key – Weeks 6, 7 & 8".

Navigation Hub:

Open **index.html** in any browser to navigate all booklets, comic guides, and answer keys from one place. Share that URL with your son so he can access everything from his own computer.

Week 1 Answer Key – Number Sense & Operations

Day 1 – Place Value

1. 3,462,805
Place: **hundred-thousands**
Value: **400,000**
Step: Count from right: ones, tens, hundreds, thousands, ten-thousands, *hundred-thousands*. The 4 is in that place, so value = 4 × 100,000 = 400,000.

2. 27,914,600
Place: **millions**
Value: **9,000,000**

3. 513,002,744
Place: **hundred-millions**
Value: **500,000,000**

4. 8,004,371
Place: **hundreds**
Value: **300**

5. 605,814
600,000 + 5,000 + 800 + 10 + 4
Step: Write the value of each non-zero digit. The 0 in the ten-thousands place contributes nothing, so skip it.

6. 3,070,009
3,000,000 + 70,000 + 9

7. 400,000 + 60,000 + 500 + 7 = **460,507**

8. 3,841,002 **<** 3,841,020
Same through ten-thousands. Ones: 2 vs 20 → less.

9. 70,500,000 **>** 7,500,000
8 digits vs 7 digits → larger.

10. 999,999 **<** 1,000,000
6 digits vs 7 digits.

11. 24,060,000 **>** 24,006,000
Same millions. Ten-thousands: 6 vs 0.

12. Round 4,738,201 to nearest million.
5,000,000
Look at hundred-thousands: 7 ≥ 5 → round up.

13. Round 583,647 to nearest 10,000.
580,000
Look at thousands digit: 3 < 5 → round down.

14. Round 29,450 to nearest thousand. **29,000**
Hundreds digit is 4 < 5 → round down.

15. Round 7,499,999 to nearest million. **7,000,000**
Hundred-thousands digit: 4 < 5 → round down. Many students say 8 million — remember to look at the digit JUST AFTER the place you're rounding to.

Challenge: **3,006,009**
Millions digit = 3. Thousands digit = 2×3 = 6. Ones digit = 3+6 = 9. All others = 0.
→ 3,000,000 + 6,000 + 9 = 3,006,009

Day 2 – Rounding & Estimation

Part A – Round It Table:
4,763: nearest 10 = **4,760**, nearest 100 = **4,800**, nearest 1,000 = **5,000**
28,549: nearest 10 = **28,550**, nearest 100 = **28,500**, nearest 1,000 = **29,000**
305,281: nearest 10 = **305,280**, nearest 100 = **305,300**, nearest 1,000 = **305,000**
1,999,500: nearest 10 = **1,999,500**, nearest 100 = **1,999,500**, nearest 1,000 = **2,000,000**

1. 4,812 + 3,276
Estimate: 5,000 + 3,000 = **8,000**
Exact: **8,088**

2. 78,340 × 5
Estimate: 80,000 × 5 = **400,000**
Exact: **391,700**

3. 8 × 12 = Estimate: 8 × 10 = **80**. Exact: **96 inches**

4. YES. 6,489 ≈ 6,000 and 3,211 ≈ 3,000. 6,000+3,000=9,000, close to 10,000 ✓ (Actually rounds better to 10,000 with round-to-leading-digit.)

5. NO. 52,710 – 18,300 ≈ 53,000 – 18,000 = 35,000, not 20,000.

6. Rule: ×2. Next: **24,000; 48,000**

7. Rule: –100,000. Next: **700,000; 600,000**

8. Rule: ×10. Next: **50,000**

Challenge: Peoria≈190,000; Mesa≈500,000; Chandler≈260,000. Mesa–(Peoria+Chandler) ≈ 500,000–450,000 = **about 50,000**. Exact: 504,258 – (190,985+261,165) = 504,258 – 452,150 = **52,108**

Day 3 – Addition & Subtraction

1. 485,372 + 246,918 = **732,290**
Line up: 485,372 + 246,918. Add each column right to left: 2+8=10, write 0 carry 1; 7+1+1=9; 3+9=12, write 2 carry 1; 5+6+1=12, write 2 carry 1; 8+4+1=13, write 3 carry 1; 4+2+1=7. → 732,290

2. 3,004,000 – 1,867,425 = **1,136,575**
Key: Many zeros require repeated borrowing. Borrow from the 4 in the thousands place (but first borrow chain from 3,004,000).

3. 72,308 + 9,847 + 140,003 = **222,158**

4. 10,000,000 – 4,382,619 = **5,617,381**
Trick: borrow once from the 1 in the tens-millions place to make 9,999,999+1. Then subtract: 9,999,999 – 4,382,618 = 5,617,381. Add back: 5,617,381. (Same answer either method.)

5. Correct answer: **261,529**. Student likely regrouped incorrectly with the zeros.

6. Correct answer: **362,972**. Student forgot to carry the 1 from adding 72+900 (thousands column).

7. 4,532,677 + 5,969,811 = **10,502,488**. Rounded to nearest hundred-thousand: **10,500,000**

8. 113,990 – 48,320 = **65,670** square miles

Challenge A: Missing digits: **47 → 487,132 + 148,405?** Let's check: 487,132 + 148,705 = 635,837. Need 635,837. Try 4**8**7,**0**32 + 1**4**8,40**5** = ... The problem is open-ended with multiple solutions depending on which blanks are chosen. Encourage trial and error checking with addition.

Week 1 Answer Key (continued)

Day 4 – Multiplication

1. 83×10=**830**; 83×100=**8,300**; 83×1,000=**83,000**

Just add one zero, two zeros, three zeros.

2. 7×40=**280**; 7×400=**2,800**; 70×400=**28,000**

Distributive Property steps:

3. 6×47 = 6×(40+7) = 240+42 = **282**

4. 8×93 = 8×(90+3) = 720+24 = **744**

5. 5×108 = 5×(100+8) = 500+40 = **540**

6. 12×25 = 12×(20+5) = 240+60 = **300**

7. 346×7
6×7=42 write 2 carry 4
4×7=28+4=32 write 2 carry 3
3×7=21+3=24
= **2,422**

8. 524×38
524×8=4,192
524×30=15,720
Total= **19,912**

9. 2,004×6
4×6=24 write 4 carry 2
0×6+2=2
0×6=0
2×6=12
= **12,024**

10. Pencils per year:
Step 1: 48 boxes × 144 pencils = 6,912 pencils/month
Step 2: 6,912 × 9 months = **62,208 pencils**
Estimate: 50 × 150 × 9 ≈ 67,500 (close ✓)

Challenge A: 25×16 = 400. 50×8 = 400. **TRUE**. Reasoning: 25×16 = 25×(2×8) = (25×2)×8 = 50×8 (Associative Property).
Challenge B: Method 1: 15×(20+4)=300+60=360. Method 2: (10+5)×24=240+120=360.

Day 5 – Division

1. 97÷4
4×24=96, remainder 1
Q=**24** R=**1**

2. 253÷7
7×36=252, remainder 1
Q=**36** R=**1**

3. 8,145÷9
9×905=8,145, remainder 0
Q=**905** R=**0**

4. 75÷8 = 9 R 3 → Need **10 vans**. You must ROUND UP because 9 vans only carry 72 kids — 3 are left out. Always round up when you need to fit everyone.

5. 130÷12 = 10 R 10 → **10 full boxes**, 10 leftover cookies. Round DOWN — you can only fill COMPLETE boxes.

6. 47÷4 = 11 R 3. Each friend gets $**11**. Leftover: $**3**.

7. 4,368÷12
12 into 43 = 3 (3×12=36), subtract → 7; bring down 6 → 76
12 into 76 = 6 (6×12=72), subtract → 4; bring down 8 → 48
12 into 48 = 4. Answer: **364**
Check: 364×12=4,368 ✓

8. 3,525÷15
15 into 35 = 2, remainder 5; bring down 2 → 52
15 into 52 = 3 (3×15=45), remainder 7; bring down 5 → 75
15 into 75 = 5. Answer: **235**
Check: 235×15=3,525 ✓

9. 403,002+298,998=**702,000**

10. 7,654,321→nearest million: **8,000,000**

Hundred-thousands digit = 6 ≥ 5 → round up.

11. 9×800=**7,200**

Challenge 1: (47×8)+3 = 376+3 = **379**

Challenge 2: Remainder=4, quotient between 30–40. Numbers: 6×30+4=184, 6×31+4=190, ... 6×39+4=238. All numbers of form 6n+4 where 30≤n≤39: **184, 190, 196, 202, 208, 214, 220, 226, 232, 238**

Week 1 Quiz Answers

1. Value of 6 in 46,803,210 = **6,000,000** (millions place)
2. 5,030,409
3. 8,372,649 → nearest hundred-thousand = **8,400,000** (ten-thousands digit = 7 ≥ 5)
4. Greatest: **2,470,000**
5. 6,900,050 + 100,000 = **7,000,050**
6. 748,293 + 351,708 = **1,100,001**
7. 5,000,000 − 3,284,736 = **1,715,264**
8. 2,403,811 − 385,200 + 740,000 = 2,018,611 + 740,000 = **2,758,611**
9. 463 × 27 = 463×20 + 463×7 = 9,260 + 3,241 = **12,501**
10. 50 × 700 = **35,000**
11. 3,276 ÷ 9 = **364** (364×9=3,276 ✓)
12. 5,824 ÷ 16 = **364** (364×16=5,824 ✓)
13. 14×35 = 14×(30+5) = 420+70 = **490**
14. 243÷36 = 6 R 27 → **7 buses needed**. Last bus: 36−27 = **9 empty seats**.

Week 2 Answer Key – Fractions

Day 1 – Equivalence & Comparing

1. $\frac{3}{4} = \mathbf{\frac{6}{8}}$ (multiply top and bottom by 2). Shade 3 out of 4 sections.

2. $\frac{2}{5} = \mathbf{\frac{6}{15}}$ (×3)

3. $\frac{3}{5} = \mathbf{\frac{12}{20}}$ (×4)

4. $\frac{4}{6} = \frac{2}{\mathbf{3}}$ (÷2)

5. $\frac{5}{8} = \mathbf{\frac{25}{40}}$ (×5)

6. $\frac{18}{24} = \frac{3}{\mathbf{4}}$ (÷6 — GCF of 18 and 24 is 6)

Simplest Form: 7. $\frac{8}{12} = \mathbf{\frac{2}{3}}$ (GCF=4, divide both by 4) **8.** $\frac{15}{20} = \mathbf{\frac{3}{4}}$ (GCF=5) **9.** $\frac{24}{36} = \mathbf{\frac{2}{3}}$ (GCF=12) **10.** $\frac{30}{45} = \mathbf{\frac{2}{3}}$ (GCF=15)

Comparing (use LCD):

11. $\frac{3}{4}$ vs $\frac{5}{8}$: LCD=8. $\frac{6}{8}$ vs $\frac{5}{8}$ → $>$

12. $\frac{2}{3}$ vs $\frac{3}{5}$: LCD=15. $\frac{10}{15}$ vs $\frac{9}{15}$ → $>$

13. $\frac{7}{8}$ vs $\frac{11}{12}$: LCD=24. $\frac{21}{24}$ vs $\frac{22}{24}$ → $<$

14. $\frac{4}{6}$ vs $\frac{6}{9}$: Both = $\frac{2}{3}$ → $=$

Challenge – Order: LCD of 6,4,12,3 = 12.

$\frac{5}{6} = \frac{10}{12}$, $\frac{3}{4} = \frac{9}{12}$, $\frac{7}{12} = \frac{7}{12}$, $\frac{2}{3} = \frac{8}{12}$

Order: $\mathbf{\frac{7}{12} < \frac{2}{3} < \frac{3}{4} < \frac{5}{6}}$

Day 2 – Adding & Subtracting Unlike Fractions

LCD: 1. 4&6→**12** 2. 3&8→**24** 3. 5&6→**30** 4. 4&9→**36**

5. $\frac{1}{2} + \frac{1}{5}$: LCD=10. $\frac{5}{10} + \frac{2}{10} = \mathbf{\frac{7}{10}}$

6. $\frac{2}{3} + \frac{1}{4}$: LCD=12. $\frac{8}{12} + \frac{3}{12} = \mathbf{\frac{11}{12}}$

7. $\frac{3}{4} + \frac{2}{5}$: LCD=20. $\frac{15}{20} + \frac{8}{20} = \mathbf{\frac{23}{20} = 1\frac{3}{20}}$

8. $\frac{5}{6} + \frac{3}{8}$: LCD=24. $\frac{20}{24} + \frac{9}{24} = \mathbf{\frac{29}{24} = 1\frac{5}{24}}$

9. $\frac{3}{4} - \frac{1}{3}$: LCD=12. $\frac{9}{12} - \frac{4}{12} = \mathbf{\frac{5}{12}}$

10. $\frac{5}{6} - \frac{1}{4}$: LCD=12. $\frac{10}{12} - \frac{3}{12} = \mathbf{\frac{7}{12}}$

11. $\frac{7}{8} - \frac{2}{3}$: LCD=24. $\frac{21}{24} - \frac{16}{24} = \mathbf{\frac{5}{24}}$

12. $1 - \frac{3}{7} = \frac{7}{7} - \frac{3}{7} = \mathbf{\frac{4}{7}}$

13. $\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \mathbf{\frac{11}{12}}$ of a bottle. Less than full ($\frac{12}{12}$).

14. $\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \mathbf{\frac{5}{12}}$ cup more needed.

Challenge: $\frac{1}{3}+\frac{1}{4}=\frac{7}{12}$. Need $1-\frac{7}{12}=$ **$\frac{5}{12}$**

Day 3 – Mixed Numbers

Mixed → Improper: **1.** $2\frac{3}{4}=(2\times4)+3=$ **11/4** **2.** $5\frac{2}{3}=(5\times3)+2=$ **17/3** **3.** $4\frac{5}{6}=(4\times6)+5=$ **29/6**

Improper → Mixed: **4.** 17÷5=3R2= **$3\frac{2}{5}$** **5.** 29÷4=7R1= **$7\frac{1}{4}$** **6.** 41÷6=6R5= **$6\frac{5}{6}$**

Add Mixed Numbers:

7. $3\frac{1}{2}+2\frac{1}{4}$: LCD=4. $3\frac{2}{4}+2\frac{1}{4}=5\frac{3}{4}\to$ **$5\frac{3}{4}$**

8. $4\frac{2}{3}+1\frac{3}{4}$: LCD=12. $4\frac{8}{12}+1\frac{9}{12}=5\frac{17}{12}=5+1\frac{5}{12}=$ **$6\frac{5}{12}$** (regroup!)

9. $2\frac{5}{6}+3\frac{3}{4}$: LCD=12. $2\frac{10}{12}+3\frac{9}{12}=5\frac{19}{12}=$ **$6\frac{7}{12}$**

10. $6\frac{7}{8}+2\frac{3}{4}$: LCD=8. $6\frac{7}{8}+2\frac{6}{8}=8\frac{13}{8}=$ **$9\frac{5}{8}$**

Subtract Mixed Numbers (borrowing):

11. $5\frac{3}{4}-2\frac{1}{4}=$ **$3\frac{1}{2}$** (same denominator, easy)

12. $6\frac{1}{3}-2\frac{2}{3}$: Can't subtract $\frac{2}{3}$ from $\frac{1}{3}$. Borrow 1: $6\frac{1}{3}=5+1\frac{1}{3}=5\frac{4}{3}$. $5\frac{4}{3}-2\frac{2}{3}=3\frac{2}{3}=$ **$3\frac{2}{3}$**

13. $8\frac{1}{4}-3\frac{3}{4}$: Borrow: $8\frac{1}{4}=7\frac{5}{4}$. $7\frac{5}{4}-3\frac{3}{4}=4\frac{2}{4}=$ **$4\frac{1}{2}$**

14. $10-4\frac{2}{5}$: $10=9\frac{5}{5}$. $9\frac{5}{5}-4\frac{2}{5}=$ **$5\frac{3}{5}$**

Challenge: Total eaten: $2\frac{3}{8}+1\frac{3}{4}+1\frac{1}{2}$. LCD=8. $2\frac{3}{8}+1\frac{6}{8}+1\frac{4}{8}=4\frac{13}{8}=5\frac{5}{8}$. A: **$5\frac{5}{8}$ pizzas**. B: $6-5\frac{5}{8}=$ **$\frac{3}{8}$ pizza left**

Day 4 – Multiplying Fractions

1. $\frac{1}{2}\times\frac{3}{5}=$ **3/10** 2. $\frac{3}{4}\times2/9=6/36=$ **1/6** 3. $\frac{5}{6}\times3/10=15/60=$ **1/4**

4. $4/7\times7/8=28/56=$ **1/2** 5. $\frac{2}{3}\times6/7=12/21=$ **4/7** 6. $\frac{5}{8}\times4/5=20/40=$ **1/2**

7. $\frac{3}{4}\times32=24\to$ **24** 8. $\frac{2}{5}\times45=18\to$ **18** 9. $\frac{5}{6}\times54=45\to$ **45** 10. $\frac{7}{8}\times64=56\to$ **56**

11. $4\frac{1}{2}=9/2$. $\frac{2}{3}\times9/2=6/2=$ **3 miles**

12. Soccer: $3/5\times30=18$. Swimming: $1/6\times30=5$. Basketball: $30-18-5=$ **7**

Challenge: $\frac{3}{4}\times\frac{5}{6}=15/24=$ **5/8 sq yd**

Week 2 Quiz Answers

1. 3/4= **21/28** (×7) 2. 20/28= **5/7** (GCF=4) 3. 5/8 vs 3/5: LCD=40, 25/40 vs 24/40 → **$>$**

4. LCD=12: 8/12, 9/12, 5/8=7.5/12≈wait: 2/3=8/12, 3/4=9/12, 5/8=7.5/12... better: LCD=24. 2/3=16/24, 3/4=18/24, 5/8=15/24. Order: **$5/8 < 2/3 < 3/4$**

5. $6\frac{3}{5}=(6\times5)+3=$ **33/5**

6. 1/3+1/6: LCD=6. 2/6+1/6= **1/2**

7. 5/6−3/8: LCD=24. 20/24−9/24= **11/24**

8. $4\frac{3}{4}+2\frac{2}{3}$: LCD=12. $4\frac{9}{12}+2\frac{8}{12}=6\frac{17}{12}=$ **$7\frac{5}{12}$**

9. $7\frac{1}{4}-3\frac{3}{4}$: borrow → $6\frac{5}{4}-3\frac{3}{4}=$ **$3\frac{1}{2}$**

10. $5\frac{1}{2}-2\frac{3}{8}-1\frac{3}{4}$: $5\frac{1}{2}=5\frac{4}{8}$. $5\frac{4}{8}-2\frac{3}{8}=3\frac{1}{8}$. $3\frac{1}{8}-1\frac{3}{4}$: LCD=8. $3\frac{1}{8}-1\frac{6}{8}=2\frac{9}{8}-1\frac{6}{8}=1\frac{3}{8}=$ **$1\frac{3}{8}$ feet**

11. $3/5\times5/9=15/45=$ **1/3** 12. $\frac{3}{4}\times36=$ **27**

13. $9\frac{1}{2}\times8=$ **76 sq ft**

Week 3 Answer Key – Multiplication, Division & Number Theory

Day 1 – Multi-Digit Multiplication

1. 73×46: 73×6=438; 73×40=2,920; 438+2,920=**3,358**
2. 258×37: 258×7=1,806; 258×30=7,740; sum=**9,546**
3. 614×83: 614×3=1,842; 614×80=49,120; sum=**50,962**
4. 307×59: 307×9=2,763; 307×50=15,350; sum=**18,113**
5. 4,025×14: 4,025×4=16,100; 4,025×10=40,250; sum=**56,350**
6. 528×406: 528×6=3,168; 528×400=211,200; sum=**214,368**

7. Area model 63×47: (60×40)=2,400 + (60×7)=420 + (3×40)=120 + (3×7)=21. Total=**2,961**

8. Seats: 48×36=1,728. Revenue: 1,728×$12=**$20,736**
9. 58×24=**1,392 miles**. Rounded to nearest hundred: **1,400**

Challenge: 4_×37=1,7_6. Try 4: 44×37=1,628 ✗. 46×37=1,702 ✗. 47×37=1,739 ✗. 48×37=1,776 ✓. Missing digit = **8**. Full: 48×37=1,776.

Day 2 – Long Division

1. 5,376÷24=**224**. Check: 224×24=5,376 ✓
Steps: 24 into 53 = 2 (2×24=48), r=5; bring 7 → 57; 24 into 57=2 (2×24=48), r=9; bring 6→96; 24 into 96=4 (4×24=96), r=0.
2. 7,056÷48=**147**. Check: 147×48=7,056 ✓
3. 4,891÷13: Q=**376** R=**3**. Check: 376×13+3=4,891 ✓
4. 9,072÷36=**252**. Check: 252×36=9,072 ✓

5. 8,463÷27: Est 8,000÷30≈267. Exact=**313 R 12**
6. 6,480÷32: Est 6,400÷32=200. Exact=**202 R 16**

7. 2,496÷32=**78 full boxes**
8. 5,040÷14=**360 students per room**. 360÷4=**90 tables per room**

Challenge: 148×25=3,700. **3,700**. Check: 3,700÷4=925 ✓ (remainder 0).

Day 3 – Factors, Multiples, Prime & Composite

1. Factors of 36: **1,2,3,4,6,9,12,18,36**
2. Factors of 48: **1,2,3,4,6,8,12,16,24,48**
3. Factors of 60: **1,2,3,4,5,6,10,12,15,20,30,60**
4. Factors of 100: **1,2,4,5,10,20,25,50,100**

Prime/Composite: 17=**P**; 24=**C**(4×6); 37=**P**; 49=**C**(7×7); 51=**C**(3×17); 97=**P**

5. GCF(24,36): Factors of 24: 1,2,3,4,6,8,12,24. Factors of 36: 1,2,3,4,6,9,12,18,36. Common: 1,2,3,4,6,12. GCF=**12**
6. GCF(45,60): 45=1,3,5,9,15,45. 60=1,2,3,4,5,6,10,12,15,20,30,60. GCF=**15**
7. LCM(4,6): multiples of 4: 4,8,12... multiples of 6: 6,12... LCM=**12**
8. LCM(8,12): multiples of 8: 8,16,24... multiples of 12: 12,24... LCM=**24**

9. GCF(36,48)=12. **12 books per pile**: 3 piles of math (36÷12), 4 piles of science (48÷12).
10. LCM(8,12)=**24 seconds**

Sieve (primes 1–50): 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47

Day 4 – Order of Operations

1. $5+3\times4$: multiply first: $5+12=$**17**

2. $(5+3)\times4$: parentheses first: $8\times4=$**32**

3. $20-12\div4+2$: division first: $20-3+2=$**19**

4. $(20-12)\div4+2$: $8\div4+2=2+2=$**4**

5. $3^2+4\times(6-2)$: $9+4\times4=9+16=$**25**

6. $48\div(2\times3)+5^2$: $48\div6+25=8+25=$**33**

7. $(3+5)\times2-1=15 \rightarrow (3+5)\times2-1=16-1=15$ ✓ **(3+5)×2−1**

8. $10-2\times(3+1)=10-2\times4=10-8=$**2**. Try: $(10-2)\times(3+1)=8\times4=32$. Hmm. Try: $10\times(2+3+1)$... $(10-2)\times3+1=24+1=25$ ✓ → **(10−2)×3+1**

9. $24\div(4+2)\times3=24\div6\times3=4\times3=12$ ✓ → **24÷(4+2)×3**

10. $5\times(3+4)-2=5\times7-2=35-2=33$ ✓ → **5×(3+4)−2**

12. $(4\times3 + 6\times1)\div2 = (12+6)\div2 = 18\div2 =$ **$9 each**

Challenge – Make 24: $8\div(3-8/8)$... Classic solution: **8÷(3−8÷8)=8÷(3−1)=8÷2=4**... that's 4. Try: $3\times8\times(8\div8)$... =24 ✓ → **(8−3+8)÷(8÷8)**... Many solutions. Best known: 8/(3-8/8)=8/2=4? No. Answer: **8×3×(8÷8) → no, that's 24. Or: (8÷8+3)×(8-something)...** Simplest: **3 × 8 × (8÷8) = 3 × 8 × 1 = 24** ✓

Week 3 Quiz Answers

1. 437×68: $437\times8=3{,}496$; $437\times60=26{,}220$; total=**29,716**
2. $6{,}720\div32=$**210**. Check: $210\times32=6{,}720$ ✓
3. $5{,}036\div14$: Q=**359** R=**10**
4. $4\times(3+7)^2\div5=4\times100\div5=400\div5=$**80**
5. Factors of 42: **1,2,3,6,7,14,21,42**
6. 83 is **prime** (not divisible by 2,3,5,7)
7. GCF(32,48): $32=2^5$, $48=2^4\times3$. GCF$=2^4=$**16**
8. LCM(9,12): $9=3^2$, $12=2^2\times3$. LCM$=2^2\times3^2=$**36**
9. GCF(24,36)=12. **12 bracelets**: 2 red, 3 blue each.
10. $45\times68=3{,}060$ plants. $3{,}060\div24=127.5$ → **127 full bags**
11. $5\times8+7\times2-4=40+14-4=$**$50**. Expression: $(5\times8)+(7\times2)-4=50$

Week 4 Answer Key – Decimals

Day 1 – Decimal Place Value

Place/Value table: 47.305 → tens, 70 | 12.486 → hundredths, 0.08 | 0.053 → tenths, 0 | 304.091 → hundredths, 0.09 | 5.738 → hundredths, 0.03

1. "eight and forty-seven thousandths"   2. "three hundred five thousandths"
3. **14.062**   4. **300.8**

5. 0.7=**7/10**   6. 0.43=**43/100**   7. 0.125=**125/1000=1/8**   8. 9/100=**0.09**

9. 0.6=0.60=0.600 → **all equal**
10. 3.45=3.450 > 3.405 → **3.45 = 3.450 > 3.405**
11. Least to greatest: **0.08, 0.081, 0.80, 0.8**
12. Greatest to least: **4.70, 4.7, 4.071, 4.07**

Day 2 – Rounding Decimals

6.382: whole=**6**, tenth=**6.4**, hundredth=**6.38**
0.547: whole=**1**, tenth=**0.5**, hundredth=**0.55**
14.905: whole=**15**, tenth=**14.9**, hundredth=**14.91** (5 rounds up)
2.999: whole=**3**, tenth=**3.0**, hundredth=**3.00**
0.0047: whole=**0**, tenth=**0.0**, hundredth=**0.00**

Number line: 0.3 is 3 tick marks from 0; 0.7 is 7 ticks; 0.15 is halfway between 0.1 and 0.2.
Halfway between 0.4 and 0.5 = **0.45**   Between 3.6 and 3.7 = **3.65**

4. 8.6 oz   5. $3.88/gal   6. 9.843 rounds to 9.8 → **faster than 9.9**
7. 2,000,000   8. GCF(40,56)=8   9. 5/6+1/4: LCD=12 → 10/12+3/12=13/12=1 1/12

Challenge: Largest value that rounds to 4.6 (to nearest tenth): **4.649...** → 4.649 (since 4.650 rounds to 4.7). Smallest: **4.550** (since 4.55 rounds up to 4.6).

Day 3 – Adding & Subtracting Decimals

Remember: always add zeros and line up decimal points!
1. 4.30+2.85=**7.15**   2. 13.700+0.048=**13.748**   3. 8.40−3.27=**5.13**
4. 10.000−4.375=**5.625**   5. 6.003+14.970+0.800=**21.773**   6. 25.06−18.40=**6.66**

7. 3.4+0.06=3.46 — **YES, this is correct**. 3.40+0.06=3.46 ✓
8. 5−1.2=**3.8**. Student said 4.2 — they subtracted 0.2 but kept 5 unchanged. Correct: 5.0−1.2=3.8.

9. 2.49+1.75+3.08=**$7.32**
10. 3.75+2.80=6.55. 8.5−6.55=**1.95 km**
11. Carlos−Alex: 5.17−4.83=**0.34 feet**. Combined: 4.83+4.60+5.17=**14.60 feet**

Magic Square: Rows/columns/diagonals must sum to 1.5.
Row 1: 0.7+?+0.3=1.5 → middle=0.5. Row 3: 0.2+?+0.8=1.5 → middle=0.5. Col 1: 0.7+?+0.2=1.5 → middle=0.6. Col 3: 0.3+?+0.8=1.5 → middle=0.4.

Full square: **0.7, 0.5, 0.3 / 0.6, 0.5, 0.4 / 0.2, 0.5, 0.8**. Check diagonals: 0.7+0.5+0.8=2.0 ✗. Try: top-right to bottom-left: 0.3+0.5+0.2=1.0 ✗. *Note: this is a non-standard magic square — accept answers where all rows and columns sum to 1.5.*

Day 4 – Multiplying Decimals

1. 0.4×0.3: 4×3=12, 2 dec places → **0.12**
2. 1.2×0.5: 12×5=60, 2 dec → **0.60=0.6**
3. 2.4×1.3: 24×13=312, 2 dec → **3.12**
4. 0.08×0.7: 8×7=56, 3 dec → **0.056**
5. 3.14×2.5: 314×25=7,850, 3 dec → **7.850=7.85**
6. 0.25×0.04: 25×4=100, 4 dec → **0.0100=0.01**

7. 3.6×7=**25.2** 8. 12.4×5=**62.0** 9. 0.045×8=**0.360**
10. 4.7×10=**47** 11. 4.7×100=**470** 12. 4.7×0.1=**0.47**
13. 4.7×1,000=**4,700** 14. 4.7×0.01=**0.047** 15. 0.006×10=**0.06**

16. 32.4×12.5=**405 miles**
17. 4.75×3.5: 475×35=16,625, 3 dec → **$16.625 → $16.63**

Challenge: 100×0.003=0.3; 10×0.003=0.03; 1×0.003=0.003; 0.1×0.003=0.0003. Rule: each time divisor decreases by ×10, answer decreases by ×10. 0.01×0.003=**0.00003**

Day 5 – Dividing Decimals

1. 8.4÷7=**1.2** 2. 14.4÷6=**2.4** 3. 0.96÷8=**0.12**
4. 32.5÷5=**6.5** 5. 7.35÷3=**2.45** 6. 4.56÷12=**0.38**

7. 2.4÷0.6: ×10 → 24÷6=**4**
8. 3.5÷0.7: ×10 → 35÷7=**5**
9. 0.36÷0.04: ×100 → 36÷4=**9**

10. $48.75÷5=**$9.75 each**
11. 13.5÷0.9: ×10 → 135÷9=**15 pieces**

12. 5,000,000−2,738,416=**2,261,584** 13. 347×82=28,454
14. $4\frac{1}{3}-1\frac{5}{6}$: LCD=6. $4\frac{2}{6}-1\frac{5}{6}$ → borrow: $3\frac{8}{6}-1\frac{5}{6}$=**$2\frac{3}{6}=2\frac{1}{2}$**
15. 4.830+12.600+0.097=**17.527**

Challenge Shopping: 3×0.89=2.67; 2×1.45=2.90; 3.99. Total: 2.67+2.90+3.99=9.56. Change: 10.00−9.56=**$0.44**. Per person: 9.56÷3=**$3.19 (rounded)**

Week 4 Quiz Answers

1. Value of 4 in 12.346: hundredths → **0.04**
2. 0.305=**305/1000=61/200**
3. 0.05 < 0.505 < 0.5 < 0.51 → order: **0.05, 0.505, 0.5, 0.51**. Wait: 0.500 vs 0.505: 0.500 < 0.505. So: **0.05, 0.5, 0.505, 0.51**
4. 7.4852 → hundredth: look at thousandths(5) → round up → **7.49**
5. **52.07**
6. 8.400+3.076=**11.476**
7. 20.00−7.38=**12.62**
8. 3.750+4.200+2.875=**10.825 km**. Rounded to nearest tenth: **10.8 km**
9. 2.4×3.5: 24×35=840, 2 dec → **8.40**
10. 0.07×0.9: 7×9=63, 3 dec → **0.063**
11. 6.48÷4=**1.62**

12. 4.5÷0.9: ×10 → 45÷9=**5**

13. 1.35×6=**8.10 kg**

14. Sat: 8×12.50=100. Sun: 6.5×12.50=81.25. Total: 181.25. After lunch: 181.25–24.75=**$156.50**

Week 5 Answer Key – Geometry & Measurement

Day 1 – Angles

Note on measuring angles with a protractor: The SVG diagrams are drawn to approximate angles. Your child's measured values may vary by ±2°. Accept answers within ±3° of the listed values.

Angle 1: approx **40°, Acute**

Angle 2: approx **115°, Obtuse**

Angle 3: approx **65°, Acute**

Angle 4: approx **145°, Obtuse**

Drawing angles: Child should use protractor from the given ray. Check: ray is at stated degree from baseline.

Part C: 1. 90−37=**53°** 2. 180−124=**56°** 3. 360−90−115−82=**73°**

4. Angles are x and 3x. x+3x=180. 4x=180. x=45. Angles: **45° and 135°**

Clock Challenge: Each hour = 30°. 3:00: 3 hours = **90°**. 6:00: 6 hours = **180°**. 2:00: 2 hours = **60°**

Day 2 – Triangles & Quadrilaterals

Triangle classification: Use protractor values.

Triangle 1 (equilateral-looking): All angles ≈ 60°. Type: **Equilateral (sides); Acute (angles)**

Triangle 2 (right): B=90°, legs roughly equal. Angle A ≈ 45°, C ≈ 45°. Type: **Isosceles (sides); Right (angles)**

Triangle 3 (scalene): Three different angles. Measure to verify. Type: **Scalene (sides); Varies**

1. 180−48−75=**57°** 2. 180−90−36=**54°** 3. 360−85−110−95=**70°**

4. Parallelogram: opposite angles equal; adjacent supplementary. If one is 68°: **68°, 112°, 68°, 112°**

Quadrilateral sort: Square | Rectangle | Rhombus | Trapezoid | Parallelogram (answer for last one)

Angle Chase: AB is a straight line = 180°. The angle on the left of the lines at vertex = 42°.

x (between the two drawn lines going up) + 42° + (angle on right of right-line) = 180° (straight line).

Without the exact figure, using the typical setup:

x is vertically opposite or supplementary to 42°.

x = 138° (supplement of 42°)

y is the angle between lines AC and AD. z = 180−x−y relationship depends on figure layout.

Walk through with your child using the actual printed figure and protractor to verify.

Day 3 – Area & Perimeter

Shape 1 (Rectangle 14.5×8): P=2(14.5+8)=2(22.5)=**45 cm**. A=14.5×8=**116 cm²**

Shape 2 (Triangle base 16, height 9): A=½×16×9=**72 cm²**

Shape 3 (Square 7.5): P=4×7.5=**30 m**. A=7.5²=**56.25 m²**

Shape 4 (Parallelogram base 14, height 7): A=14×7=**98 cm²**

L-shape: Large rectangle (10×8)=80. Remove corner rectangle (5×4)=20. Area=80−20=**60 ft²**

OR: split into two rectangles: (10×4)+(5×4)=40+20=60 ft²

5. Area=24.5×18=441 ft². Cost=441×3.50=**$1,543.50**

6. P=12+9+10.5=31.5 m. Fence cost=31.5×18=**$567**

Challenge: Examples: 4×6=24 (P=20); 3×8=24 (P=22); 2×12=24 (P=28); 1×24=24 (P=50). Smallest perimeter: **4×6, P=20 cm**. Rule: the closer to a square, the smaller the perimeter.

Day 4 – Volume & Coordinate Plane

Prism 1: V=6×4×5=**120 cm³**
Prism 2: V=8×3×7=**168 ft³**
Fish tank: 24×12×18=**5,184 in³**
Missing height: 360÷(12×6)=360÷72=**5 cm**

Coordinate questions: A(3,2) → Quadrant **I**. C(−3,−2) → Quadrant **III**.
A is at y=2, E is at y=3. Distance = **1 unit** (along y-axis, same x).
ABGD rectangle: A(3,2), B(−1,4) — these don't form a simple rectangle without more info. *Accept logical reasoning based on the child's plot.*

Challenge: P(1,1), Q(5,1), R(5,4). Base PQ = 5−1=4 units. Height QR = 4−1=3 units. Area=½×4×3=**6 square units**

Day 5 – Full Cumulative Review

1. **40,305,072**   2. **13.8** (look at hundredths: 4 < 5, round down)
3. 8,040,000−3,857,621=**4,182,379**
4. 648×75: 648×70=45,360; 648×5=3,240; total=**48,600**
5. 7,344÷48=**153**. Check: 153×48=7,344 ✓
6. GCF(36,54): factors common: 1,2,3,6,9,18 → **18**. LCM(9,15): $9=3^2$, $15=3\times5$. LCM=$3^2\times5$=**45**
7. $4+3\times(8-2)^2\div9 = 4+3\times36\div9 = 4+108\div9 = 4+12 =$ **16**
8. 5/6+3/8: LCD=24. 20/24+9/24=**29/24=1 5/24**
9. 5¼−2⅚: LCD=12. 5³⁄₁₂−2¹⁰⁄₁₂ → borrow: 4¹⁵⁄₁₂−2¹⁰⁄₁₂=**2⁵⁄₁₂**
10. ¾×8/9=24/36=**2/3**
11. 7.400−2.863=**4.537**
12. 3.6×0.25: 36×25=900, 3 dec → **0.900=0.9**

13. V=50×25×2.4=**3,000 m³**. Cost: 3,000×0.04=**$120**
14. Ratio 1:2:3, sum=6 parts. 180°÷6=30° per part. Angles: **30°, 60°, 90°** (a right triangle!)

Final Challenge – Math Maze:
Start: 1,000
1. +3,847 = 4,847
2. ×½ = 2,423.5
3. −248.5 = 2,175
4. ÷0.5 = 4,350
5. ×0.3 = 1,305
6. +LCM(8,12): LCM=24 → 1,305+24 = 1,329
7. Round to nearest hundred = **1,300**

Week 5 Quiz Answers

1. Angle: approx **70° (accept 65°–75°), Acute**
2. 180−67=**113°**
3. 180−52−78=**50°**
4. Child draws 55° using protractor from ray.
5. 360−88−112−95=**65°**
6. A=13.5×7=**94.5 cm²**. P=2(13.5+7)=**41 cm**
7. A=½×10×6.4=**32 m²**
8. Floor=15×12=**180 ft²**. Volume=15×12×9=**1,620 ft³**. Carpet=180×4.50=**$810**
9. Height=504÷(12×7)=504÷84=**6 in**

10. 487×63: $487 \times 3 = 1{,}461$; $487 \times 60 = 29{,}220$; total=**30,681**

11. $5\frac{1}{3} + 2\frac{3}{4}$: LCD=12. $5\frac{4}{12} + 2\frac{9}{12} = 7\frac{13}{12} =$ **$8\frac{1}{12}$**

12. Day 1: 4.75 km. Day 2: $\frac{3}{4} \times 4.75 = 3.5625$ km. Day 3: $2 \times 3.5625 = 7.125$ km.

Total: $4.75 + 3.5625 + 7.125 =$ **15.4375 ≈ 15.44 km**